



Koneru Lakshmaiah Education Foundation

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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I - III Semester(2025-26)
Faculty Name	Ms. Alluri Swathi
Student Name	Sivarapu Rishil
Roll Number	2520090174
Branch	CSIT
Course Outcome	2

Case Study Topic: UrbanAI – Intelligent Smart City Infrastructure Optimization Platform using segment tree

Work Done Summary:

Implemented a Segment Tree in Java to efficiently manage and analyze smart city traffic datasets. Stored traffic data from different city zones and performed fast range query operations for urban traffic analytics. Implemented methods for building the Segment Tree, updating traffic data, and retrieving cumulative traffic information from selected city regions. The system efficiently analyzed traffic density between multiple smart city zones. Compared Segment Tree performance with normal array-based searching and observed faster query and update operations for large-scale smart city traffic management systems.

Implementation Details :

***Created Segment Tree structure for storing smart city traffic data.**

***BuildTree()** constructs the Segment Tree using traffic statistics from multiple city zones.

***RangeQuery()** calculates total traffic density between selected city regions efficiently.

***Update()** modifies traffic data dynamically when city traffic conditions change.

***TrafficAnalytics()** generates traffic reports for smart city monitoring.

***Traversal methods** display traffic statistics from different city zones.

***The main method** demonstrated tree construction, range queries, traffic updates, and smart city analytics operations.

Result : Screenshots/Output:

```
===== SMART CITY TRAFFIC DATA =====

Zone 1 Traffic Count : 120
Zone 2 Traffic Count : 250
Zone 3 Traffic Count : 180
Zone 4 Traffic Count : 300
Zone 5 Traffic Count : 220
Zone 6 Traffic Count : 150

===== TRAFFIC ANALYTICS REPORT =====

Total Traffic from Zone 2 to Zone 5 : 950

Updating Traffic Data for Zone 3...

===== SMART CITY TRAFFIC DATA =====

Zone 1 Traffic Count : 120
Zone 2 Traffic Count : 250
Zone 3 Traffic Count : 400
Zone 4 Traffic Count : 300
Zone 5 Traffic Count : 220
Zone 6 Traffic Count : 150

Updated Traffic from Zone 2 to Zone 5 : 1170

===== SMART CITY ANALYTICS COMPLETED =====
```

Time Complexity:

- *Segment Tree Build Operation: $O(n)$
- *Range Query Operation: $O(\log n)$
- *Update Operation: $O(\log n)$
- *Traffic Data Traversal: $O(n)$
- *Compared to normal array range searching $O(n)$, Segment Trees provide faster query and update operations for large-scale smart city analytics systems.

GitHub Link : https://github.com/rishill10/DSA2_CO2

Student Signature	Faculty Signature

